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Pipeline Installation on Zero Radius Bending (ZRB) Sleepers for Large Diameter Concrete Coated Pipelines

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Abstract

Lateral buckling, predominantly induced by operational temperature and pressure expansion, remains a key design constraint for large-diameter subsea pipelines. The deployment of zero-radius bending (ZRB) sleepers presents an effective strategy for controlling lateral buckling by inducing a sharp, localized bend at a predetermined pipeline location. However, the installation of heavy, stiff, and concrete-coated pipelines onto ZRB sleepers introduces substantial technical challenges, particularly in maintaining structural integrity and precise geometrical control during critical phases of operation.

This paper outlines a validated methodology for installing large-diameter concrete-coated pipelines onto ZRB buckle sleepers based on recent offshore project experience in Qatar. The developed procedure utilizes a Derrick lay barge (DB30) equipped with a truss stinger for the S-lay operations. Pre-lay analysis is used to optimize the vessel parameters and ensure persistent pipeline contact with the ZRB sleeper's bollard throughout the bend induction process. A critical process deviation from conventional practice involved achieving the required sharp bend radius through precise vessel heading adjustments directly at the ZRB post, thereby eliminating the need to drag the pipe along the seabed. The pipeline, featuring 38 inch & 28 inch diameter with 40 mm concrete weight coating over thick-walled pipe sections, was engineered to interface with the ZRB bollard at a predetermined elevation above the sleepers, thereby leveraging pipe stiffness and controlled vessel movement to maintain geometric conformity.

Post-installation surveys and analytical verification confirmed the structural integrity of the pipeline within the ZRB region, with key parameters, such as sag bend stresses and DNV LCC Unity Check values, remaining within the specified design thresholds. The top tension and touchdown point tension were systematically optimized using tailored lay tables corresponding to varying water depths and pipeline configurations. Furthermore, the installed ZRB sleepers effectively resisted installation-induced vertical and lateral hydrodynamic loads, thereby confirming the robustness of the design.

This field application substantiates a streamlined and robust methodology for installing large-diameter concrete-coated pipelines onto ZRB buckle sleepers using precise vessel positioning in place of pipe dragging for curve induction. This approach offers superior control and installation reliability, marking a notable advancement in lateral buckling mitigation for offshore pipeline construction. The resulting

improvements in operational efficiency and risk reduction are expected to enhance project performance and long-term pipeline reliability.

Introduction

The development of subsea hydrocarbon pipelines operating under high-pressure and high-temperature (HP/HT) conditions is driven by the global demand for energy resources. These elevated service conditions result in significant thermal and pressure-related expansions in offshore pipeline systems. When the axial movement is restricted by seabed friction, the resulting compressive forces can surpass the critical buckling threshold, leading to lateral pipeline displacement. Although intentional lateral buckling is occasionally employed to alleviate axial stresses, uncontrolled buckling poses critical threats, including excessive bending, local instability, potential fatigue failure, pipeline walking, and an escalated risk of hydrocarbon release into the environment. Such risks are magnified for large-diameter pipelines with substantial concrete weight coatings, which, while essential for bottom stability, add rigidity and further complicate the initiation and control of buckling.(Liu et al. 2014)

A range of industry mitigation strategies have been developed to manage lateral buckling, including the vertical upset method, snake-lay installation, sleeper configurations, buoyancy modules, and anchor systems. Each approach introduces deliberate geometric imperfections or alters the resistance to promote controlled buckling. However, the reliability of these techniques often depends on variable site conditions, such as soil-pipeline interactions, and may not consistently initiate buckles at the intended locations. The Zero-Radius Bending (ZRB) method represents a significant advancement in the management of controlled buckling. By integrating a specialized subsea sleeper and vertical post, the ZRB solution establishes a defined, sharp pipeline offset during installation, reliably triggering lateral buckling under predictable, controlled conditions, and lower axial forces during operational expansion.(Feng et al. 2015)

This paper presents a comprehensive technical analysis and offshore project case study that demonstrates the application of the ZRB methodology in managing lateral buckling for a large-diameter concrete-coated export gas pipeline. The project, executed offshore Qatar as part of a major engineering, procurement, construction, and installation (EPCI) campaign, utilized a conventional S-lay spread equipped with a DB30 Derrick lay barge to achieve a precise pipeline alignment and ZRB configuration. The aim of this study was to critically assess the technical feasibility, installation practices, and operational performance of ZRB deployment in HP/HT and large-diameter pipeline applications. This study draws upon validated analytical results, procedural details, and as-built data obtained during field operations.

The subsequent sections detail the technical principles and design methodology of the ZRB, project-specific parameters for pipelines and vessels, and the comprehensive installation sequence employed in the current case study. The technical analyses focused on stress and strain management, adherence to DNV standards, and evaluation of loading on sleepers and equipment. Additionally, the installation results are presented, highlighting the achieved bend radii and verification of pipeline integrity. This study concludes by synthesizing the lessons learned, addressing the operational challenges, and providing practical recommendations, serving as a valuable reference for practitioners considering the ZRB as a solution for managing complex offshore pipeline buckling.

ZRB Design Methodology

Zero-Radius Bending Concept

The Zero-Radius Bending concept involves forcing the pipeline to undergo an almost point-curvature bend during installation, effectively introducing a sharp lateral deflection at a predetermined location. In practice, "zero-radius" is a bit of a misnomer – it implies an infinitely small bend radius rather than literally zero. The pipeline is laid over a specialized buckle-trigger structure that comprises a low sleeper and an upright post

(or a fin). As the lay barge progresses, it deviates laterally, causing the pipe to be pulled against the post and partially wrapped around it. This creates a concentrated bend in the pipeline horizontally (around the post) and vertically (over a sleeper). Figure 1 illustrates the ZRB concept: the pipeline initially touches down straight over the sleeper, and then the vessel shifts to one side, forcing the pipe to hug the post and introducing a tight kink. This induced lateral out-of-straightness remains an initial imperfection on the seabed. When the pipeline is pressurized and heated during operation, the buckle will reliably form at this location under a much lower axial force than that required for a straight pipeline. In essence, the ZRB trigger "pre-installs" a buckle with a chosen amplitude. Multiple ZRB triggers can be spaced along a line to ensure that the expansion is shared among several small buckles instead of accumulating into a few large ones (which could overstress the pipe). The concept was initially developed to address cases in which the weld strain capacity was limited, and soil friction was uncertain, posing a risk of natural buckle initiation.

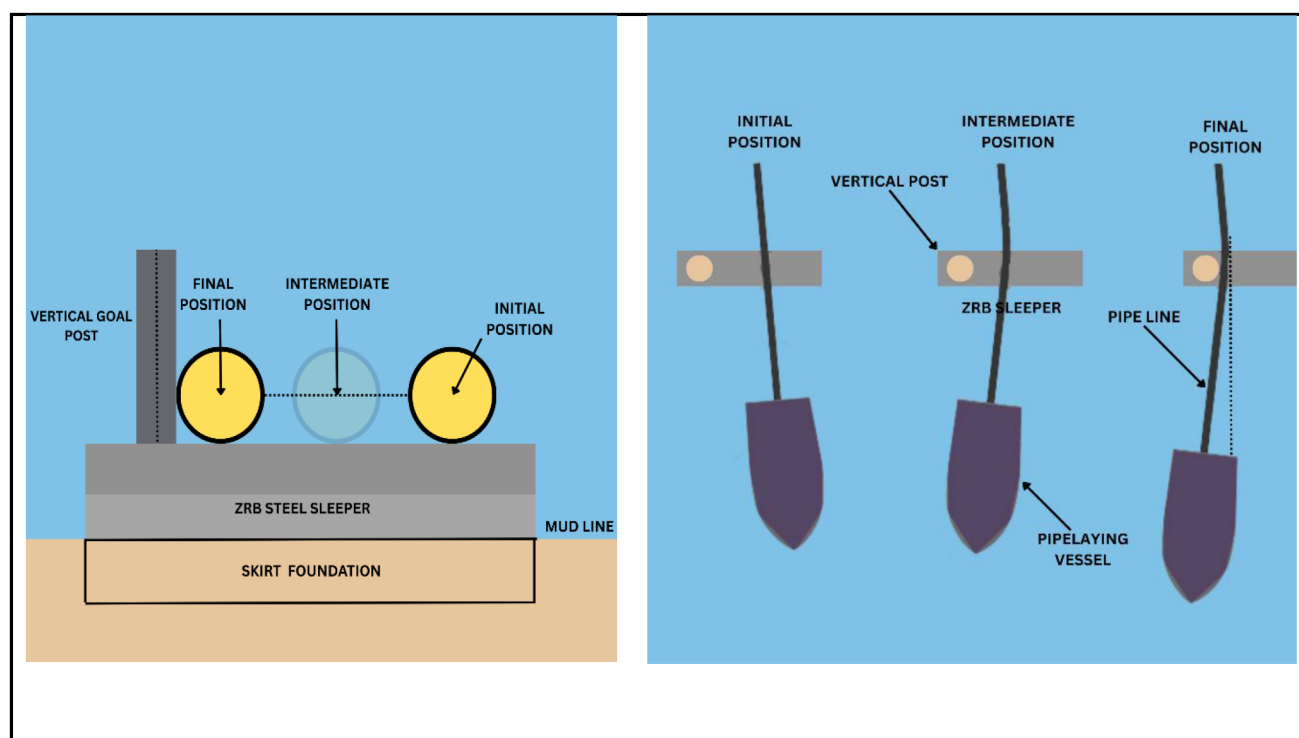


Figure 1—Schematic Sketch of a Derrick Barge installing pipeline over the ZRB Sleeper

ZRB Sleepers Design

The design of zero-radius bend (ZRB) sleepers for offshore pipeline systems is pivotal to ensuring structural integrity and operational reliability in subsea environments. These steel structures are engineered to mitigate pipeline spans and facilitate controlled buckling, thereby addressing the challenges posed by seabed irregularities and thermal expansion. Positioned strategically based on precise seabed coordinates and water depths, ZRB sleepers are designed for a 25-year service life and adhere to international standards, such as API and AISC. Advanced structural analysis software was used for modeling and code checks to ensure that the sleepers were robust, cost-effective, and optimized for safe fabrication, transportation, and installation. The design prioritizes safety, environmental resilience, and compliance with the best industry practices.

The structural design process for ZRB sleepers encompasses both in-service and pre-service conditions to ensure their performance under diverse loads. In-service analyses evaluate the stability under pipeline loads and environmental forces, including wave and current loads, calculated using established hydrodynamic equations and wave theories. The on-bottom stability and seabed bearing capacity are rigorously assessed, with foundations designed based on geotechnical data to prevent settlement or overturning. Fatigue analysis

accounts for cyclic environmental loads to ensure long-term durability, whereas corrosion is mitigated through the use of cathodic protection systems. Pre-service considerations include transportation analyses for gravity and inertia loads, as well as lifting analyses that incorporate dynamic amplification factors (DAFs). These analyses optimize the sling angles and consequence factors for safe installation. These analyses ensure that the sleepers can withstand the rigors of deployment in challenging offshore conditions.

Figure 2 shows the typical ZRB sleeper designed for this case study, which is presented in the following sections. Its approximate weight is 77 MT.

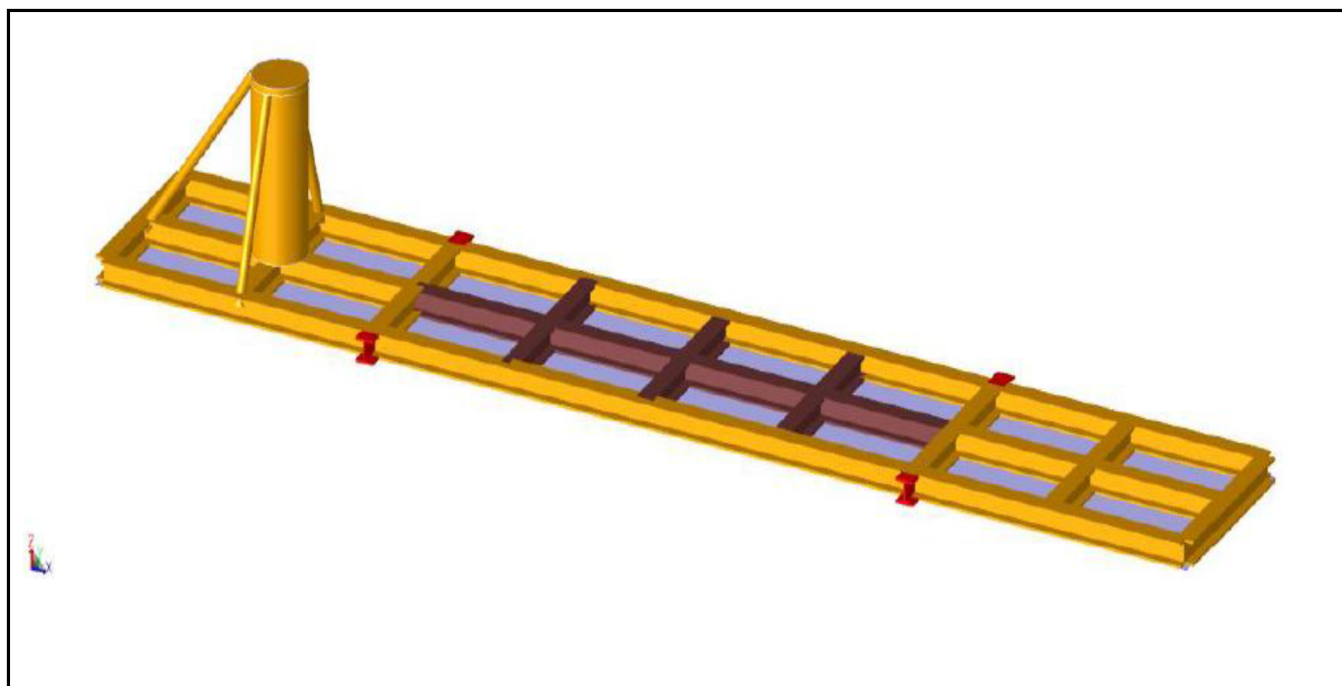


Figure 2—Typical ZRB Steel Sleeper SACS Model

ZRB Deployment

Having established the foundational design principles of ZRB sleepers, the focus now shifts to the intricacies of their deployment, a critical phase that transforms design into operational reality. "The deployment of ZRB offshore sleepers constitutes a highly sophisticated marine operation, meticulously planned to transfer these structures from a heavy-lift vessel to their precise seabed location. This process primarily utilizes specialized installation vessels, in the current case study, which have been taken from the live project, DP-2 vessels with a 300 MT/800 MT main crane capacity, selected for their robust lifting capabilities and dynamic positioning systems. The most critical phase of this operation is the "splash zone" transit, where the sleeper passes through an air-water interface. During this stage, the structure is acutely susceptible to significant dynamic loads, including slamming loads from the water impact, drag forces from movement through the water, and added mass effects. To ensure safety and feasibility, a rigorous dynamic analysis was conducted using advanced simulation software, such as Orca Flex, adhering to industry standards, including DNVGL-RP-N103: Modelling and Analysis of Marine Operations. This analysis models the sleeper as a 6D buoy and the intricate rigging system comprising spreader bars, wire ropes, and webbing slings as interconnected elements, incorporating vessel Response Amplitude Operator (RAO) data to simulate realistic motion responses. The analysis further employs irregular wave analysis, utilizing a "windowed approach" as shown in Figure 4 to capture the most critical wave conditions within a 3-hour wave train, and assesses two conservative scenarios: the structure above still water level (for slamming loads) and fully

submerged (for drag and added mass effects). A visual representation of the modeling approach is shown in Figure 3

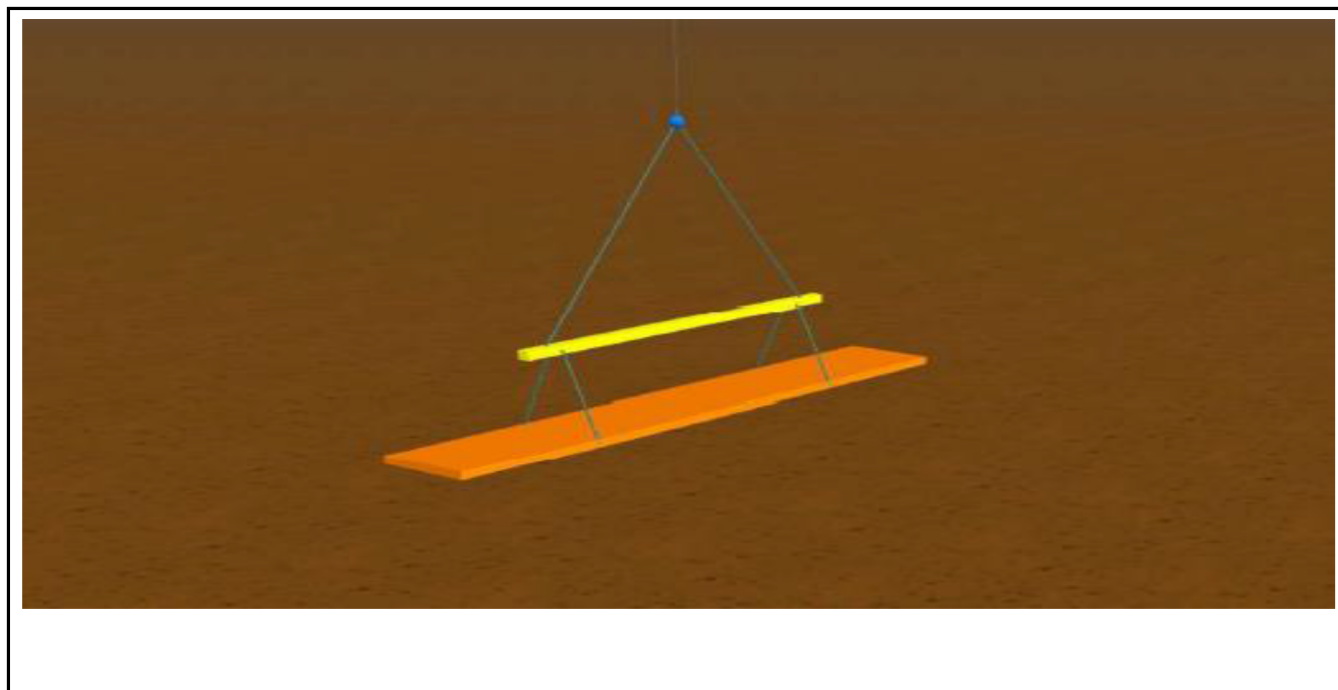


Figure 3—Typical Orca Flex Model of ZRB Sleeper Deployment

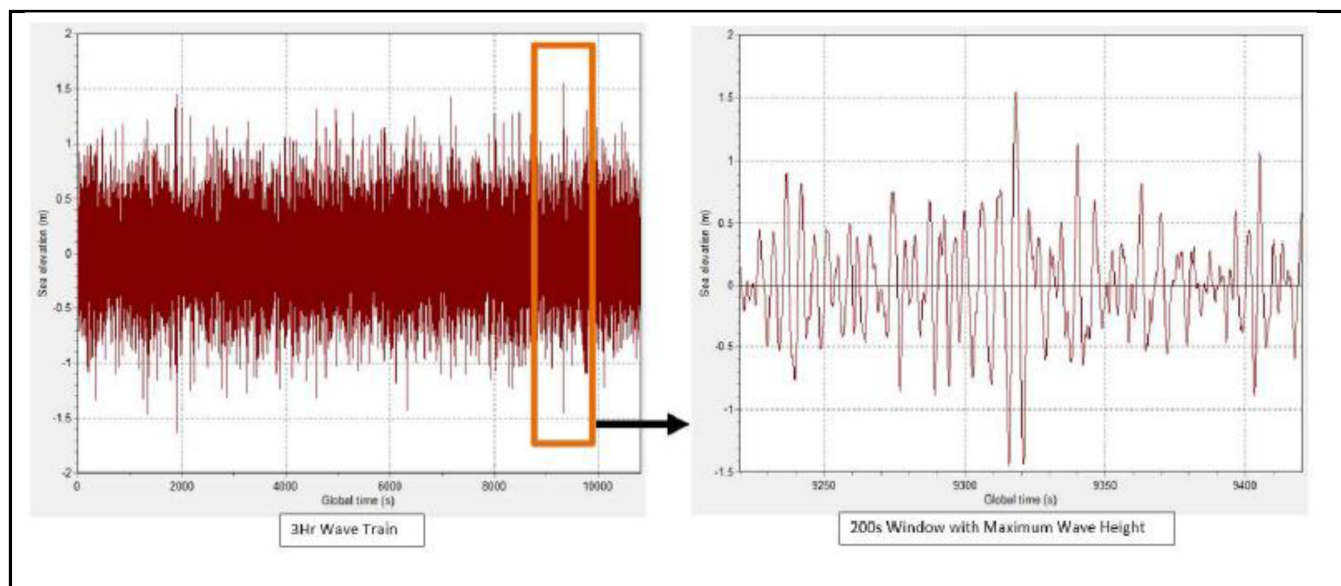


Figure 4—Windowed approach



Figure 5—Typical deployment of ZRB Sleeper Deployment

Successful deployment is contingent upon rigorous adherence to several critical considerations and best practices derived from a comprehensive engineering analysis. Operations are strictly confined to predefined allowable sea states, determined by the significant wave height (H_s) and wave direction, ensuring that the environmental forces remain within the lifting system's design capabilities. For instance, deployment may be permissible up to an H_s of 1.5 m, with specific limits varying by wave direction and vessel heading. These limits are crucial for mitigating the risks associated with excessive vessel motion and dynamic loading. The Dynamic Amplification Factor (DAF), which quantifies the amplification of static loads owing to dynamic effects, was consistently found to be less than or equal to the DAF considered in the rigging design, thus validating a conservative and robust design philosophy. Furthermore, detailed analyses confirmed that the maximum dynamic loads on the crane hook and individual rigging components remained well within their Safe Working Load (SWL) limits, with no observed slack in the rigging throughout the deployment. This ensures continuous control and prevents the occurrence of uncontrolled movements. [Figure 6: Limiting Sea States for ZRB Sleeper Deployment by Vessel Type](#) illustrates the operational envelopes, while [Table 1](#) and [Table 2: Comparison of Dynamic Amplification Factors \(DAF\) for ZRB Sleeper Deployment](#). The selection of installation vessels was based on their proven capability to handle the specific weights and dimensions of the ZRB sleepers. Operational discretion by the Vessel Master or Barge Superintendent may allow real-time adjustments based on equipment performance and actual vessel motion.

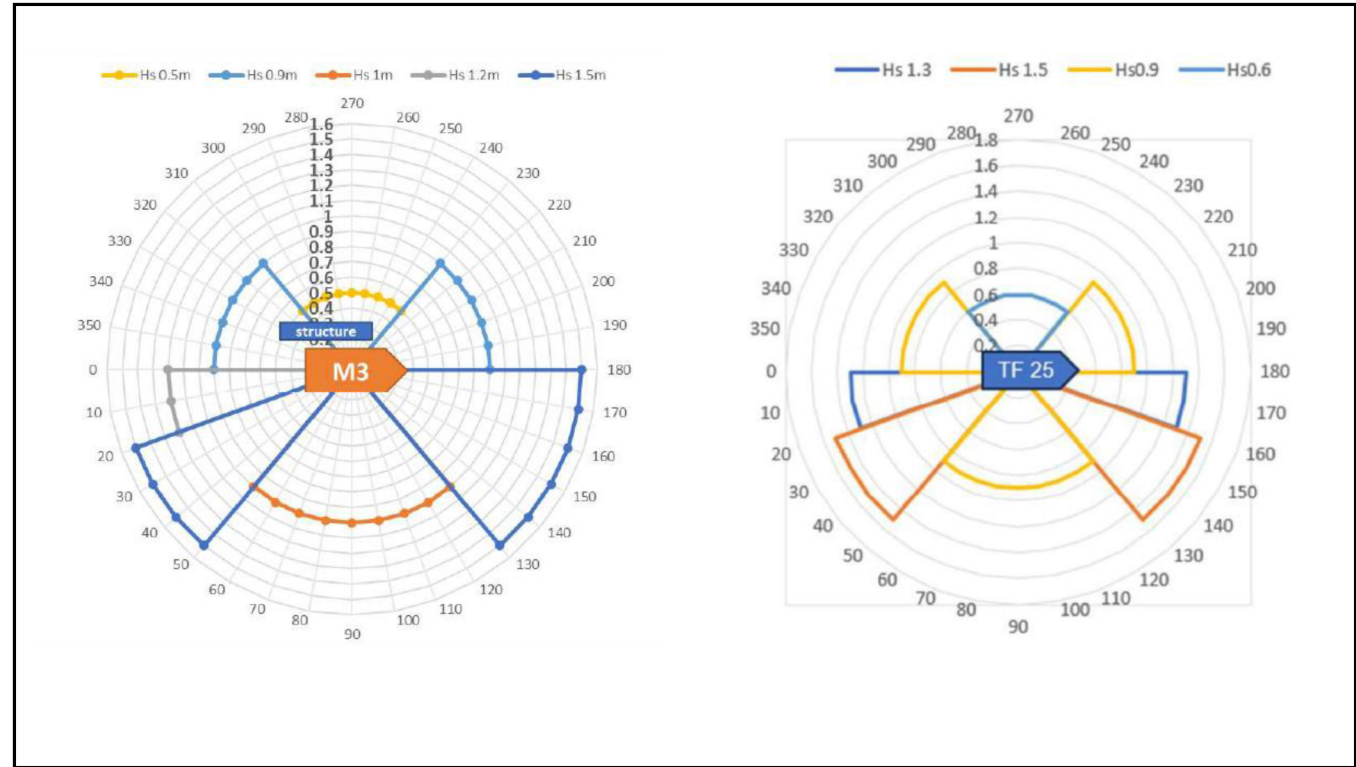


Figure 6—Limiting Sea States

Table 1—DAF for the Development of the ZRB Sleepers (DP-2 Vessel 1)

ZRB sleepers	DAF Deployment Analysis	DAF Rigging Analysis
1-01 A	1.33	1.351
1-02B	1.28	1.353

Table 2—DAF for the Development of the ZRB Sleepers (DP-2 vessel 2)

ZRB sleepers	DAF Deployment Analysis	DAF Rigging Analysis
1-01 A	1.30	1.351
1-02B	1.34	1.353

Case Study

This study offers a comprehensive case study of an ongoing Engineering, Procurement, Construction, and Installation (EPCI) project. It highlights the installation of 38-inch trunk lines and 28-inch intra-field lines, which have been implemented at varying water depths using different types of ZRB sleepers in the North Field Offshore area of Qatar. Further key details and insights are explored in the subsequent sections.

Design data

The essential characteristics of the 38-inch trunk lines and 28-inch intra-field lines are presented in Table 3. Furthermore, four distinct types of ZRB sleepers were used for the installation of these pipelines, with their specifications detailed in Table 4.

Table 3—Pipeline design data

Pipeline [Description]	Pipe OD [in]	Pipe WT [mm]	CWC thickness [mm]	WD [m]	
				Min	Max
28" pipeline (Intra-Field Lines)	28	25.4	40	21	58
38" pipeline (Trunklines)	38	26.97	40	18	52

Table 4—ZRB sleepers design data

Type of ZRB sleepers	Dimension (m)			Weight in Air	Weight in Water
	L	W	H	[MT]	[MT]
1-01A	20	3	0.37	58.50	42.26
1-01B	20	4	0.37	73.22	52.46
1-02A	20	3	1.00	62.97	46.03
1-02B	20	4	1.00	77.91	56.40

ZRB configuration for Trunklines:

For all trunk lines, the 38-inch pipeline was installed over a single ZRB sleeper, measuring approximately $20 \times 4 \times 0.37/24 \times 3 \times 0.37$ m. The pipeline's heading will remain consistent until it reaches the seabed, which is located approximately 40 m from the sleeper. At this juncture, a change in heading occurs as the vessel shifts its position transversely. Figures 8 and 9 depict the typical ZRB configuration for the 38-inch trunk lines.

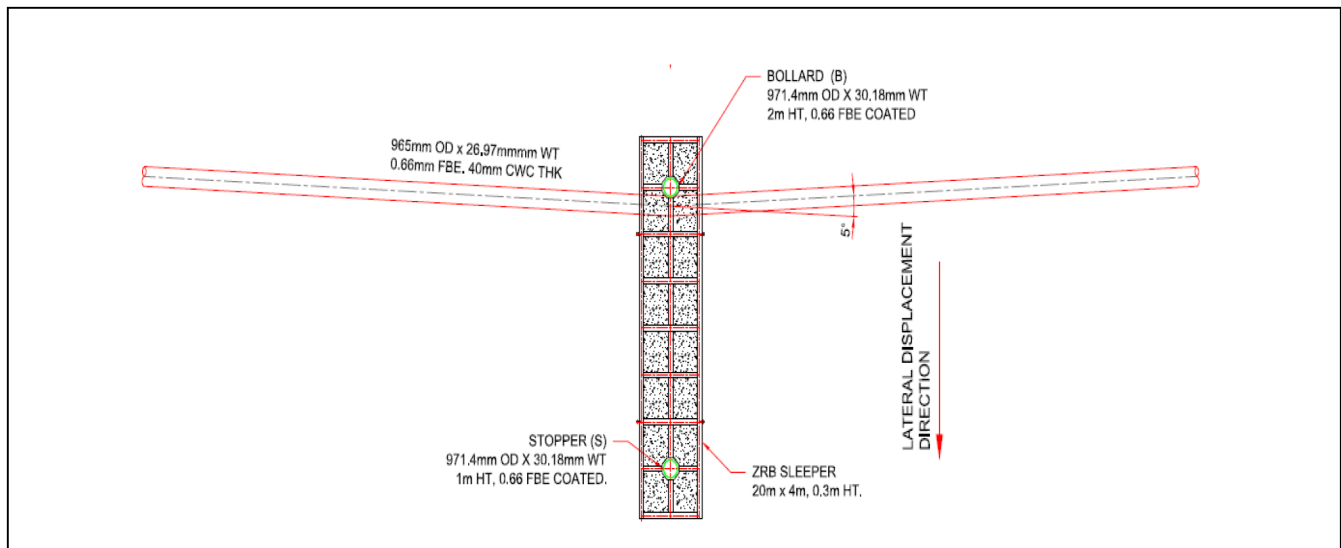


Figure 8—Typical Detailed ZRB Configuration for Trunklines

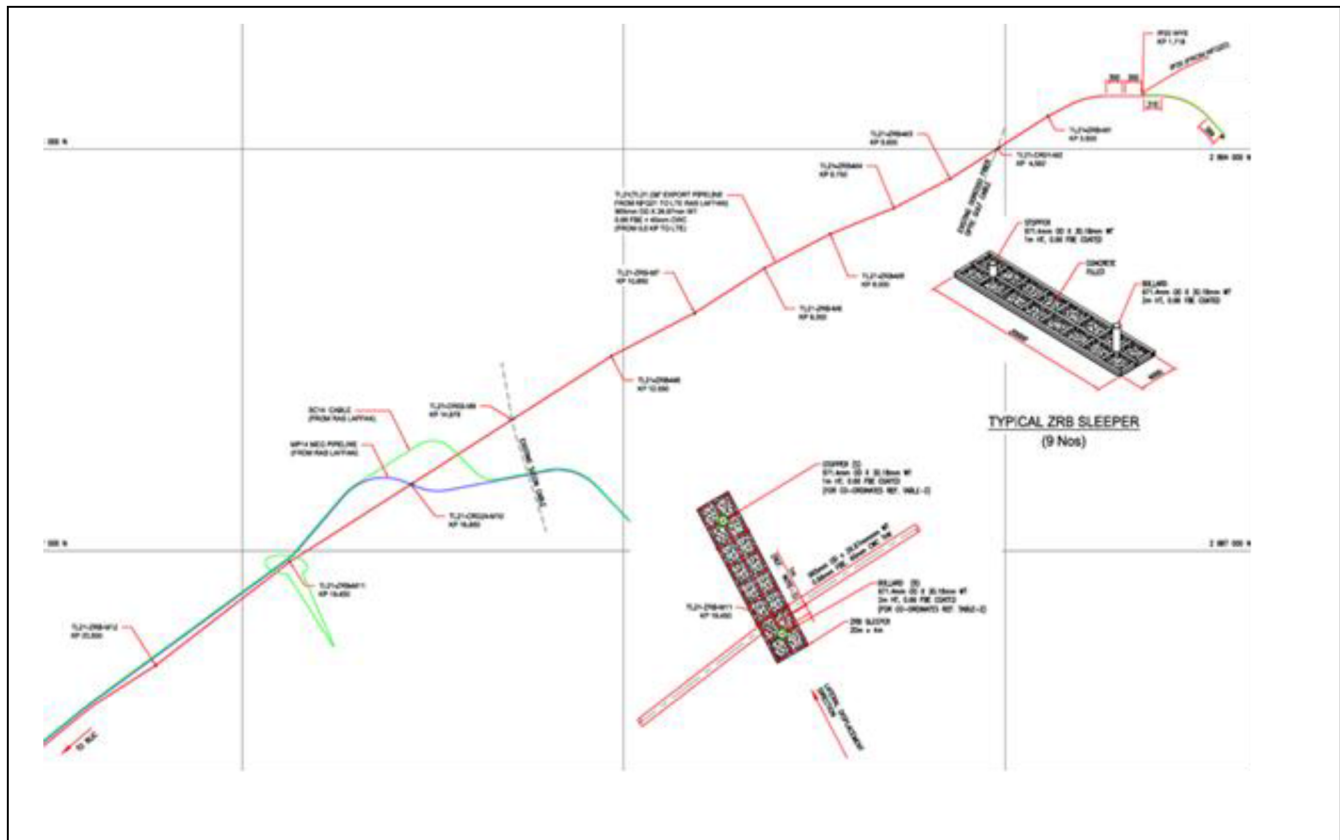


Figure 9—Typical ZRB Configuration for Trunklines

ZRB Configurations for Intra-Field Pipelines

Intra-field lines are laid straight over the first 0.4-meter height steel sleeper, and then to the 1-meter height ZRB sleeper (Bollard sleeper), and were extended to the second 0.4-meter height steel sleeper. The vessel heading remains constant until it achieves touchdown with the second 0.4-meter height sleeper. At this point, the Derrick barge changed its heading.

These configurations are applicable to the ZRB trigger location, where there is a nominal pipeline change-of-heading angle of 5° , and for route curves where the ZRB sleeper is located near the intersection points of the curves. Figures 10 and 11 depict the typical ZRB configuration for the 28-inch intra-field lines.

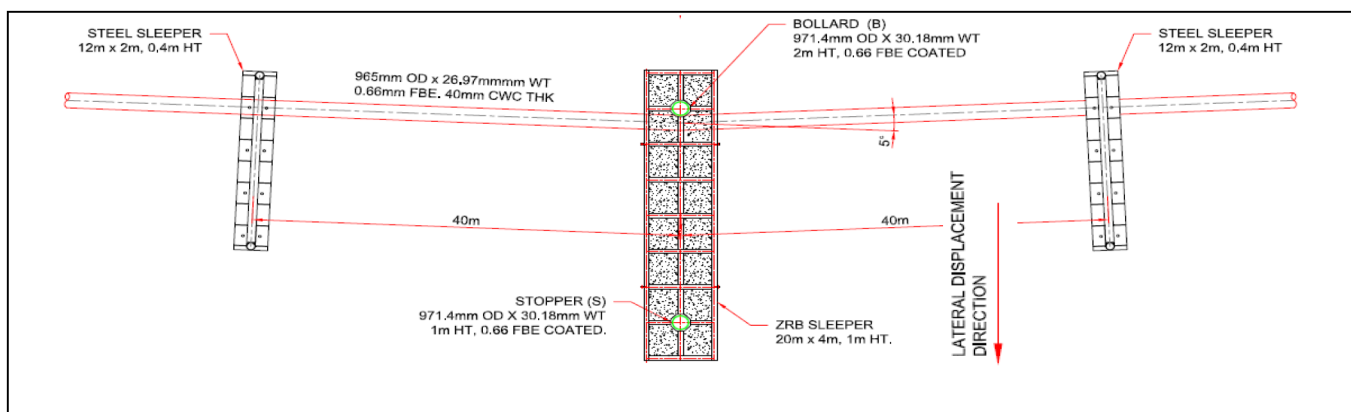


Figure 10—Typical Detailed ZRB Configuration for Intra-field pipelines

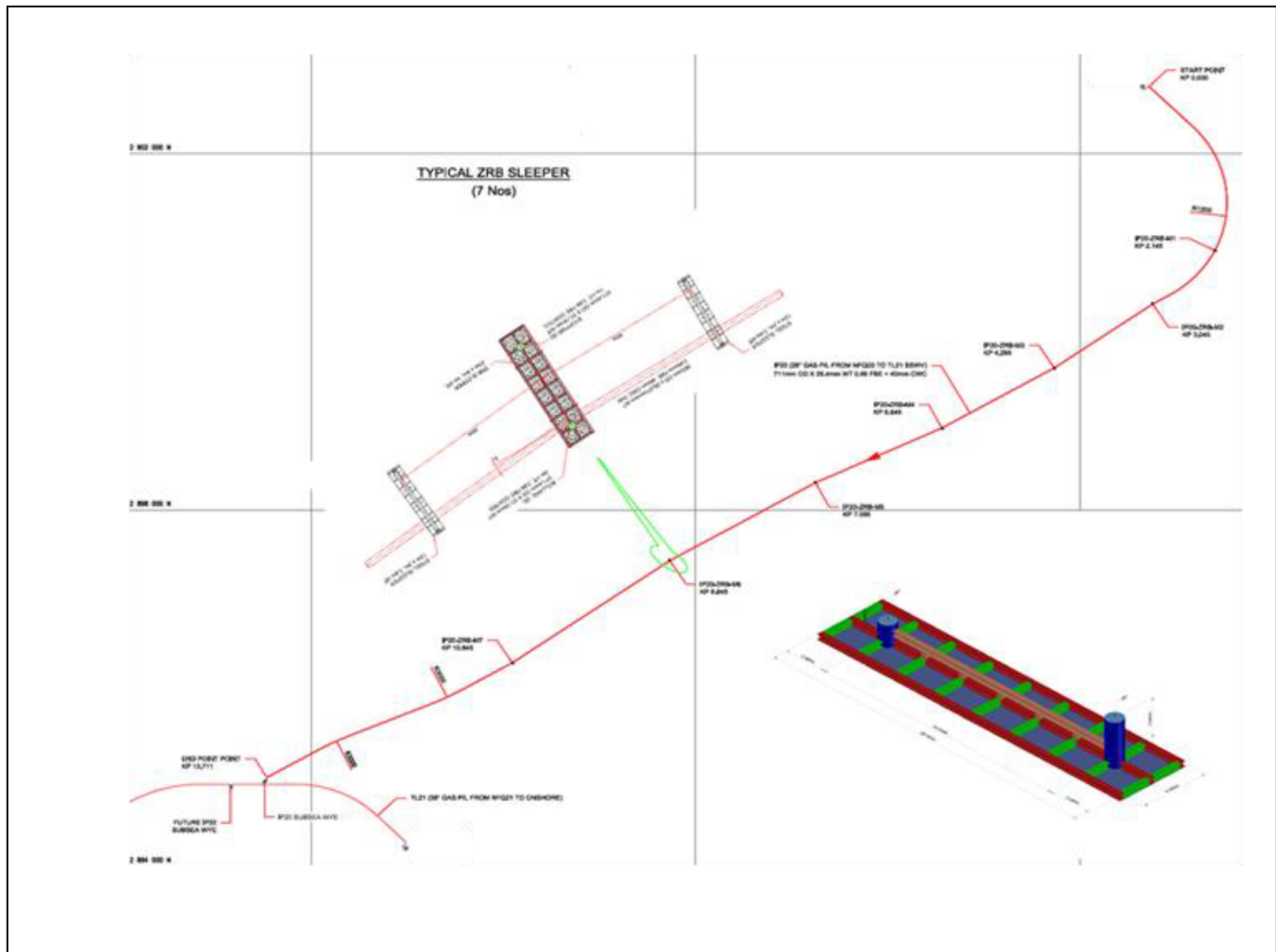


Figure 11—Typical ZRB Configuration for Intra-field Pipelines

ZRB Pipe-Lay-Results

A thorough dynamic analysis of the pipeline installation process over the ZRB sleepers was conducted using the OrcaFlex software suite. The design sea state considered a significant wave height of 1.5 m from the northwest, focusing on the vessel's response in beam sea conditions, which are known to present the most critical case.

To ensure the feasibility of the ZRB installation in relation to the pipeline reaction forces on the stinger rollers and the ZRB sleeper bollard, a comprehensive series of sensitivity analysis simulations was performed based on the following configuration:

- a. Size and WT: 28" OD / 25.4mm WT, 40mm CWC & 38" OD / 26.97mm WT, 40mm CWC
- b. Water Depth and Lay Tension:
 - i. 28" OD: 21mWD/50MT & 58mWD/90mT
 - ii. 38" OD: 18mWD/35mT & 52mWD/55mT
- c. Type of Stinger (simplified lateral friction of 0.5)
 - i. Rigid truss stinger
- d. Structural 5% damping vs. no damping (simplified lateral friction of 0.5)

The figures below depict that the white color of the pipeline section represents the part in contact with the seabed. Similarly, the pipelay process for intra-field pipelines (IPs) over ZRB sleepers and adjacent steel sleepers is illustrated in the images below.

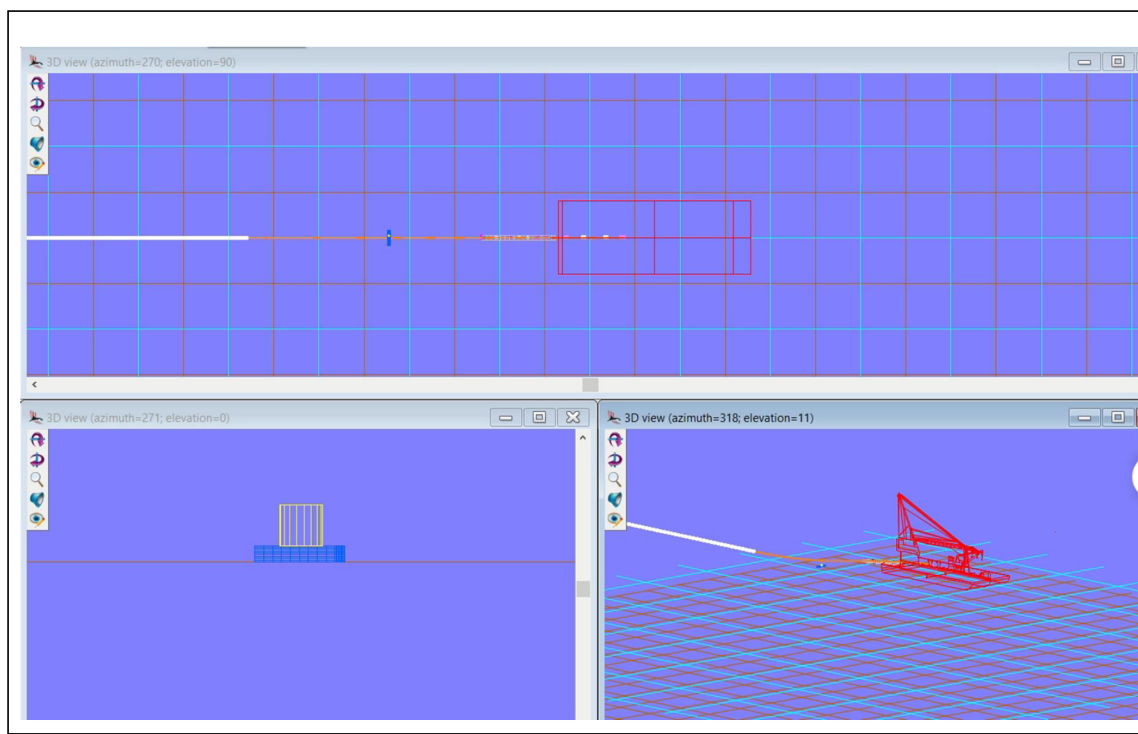


Figure 12—Typical Trunkline TDP approaching Sleeper

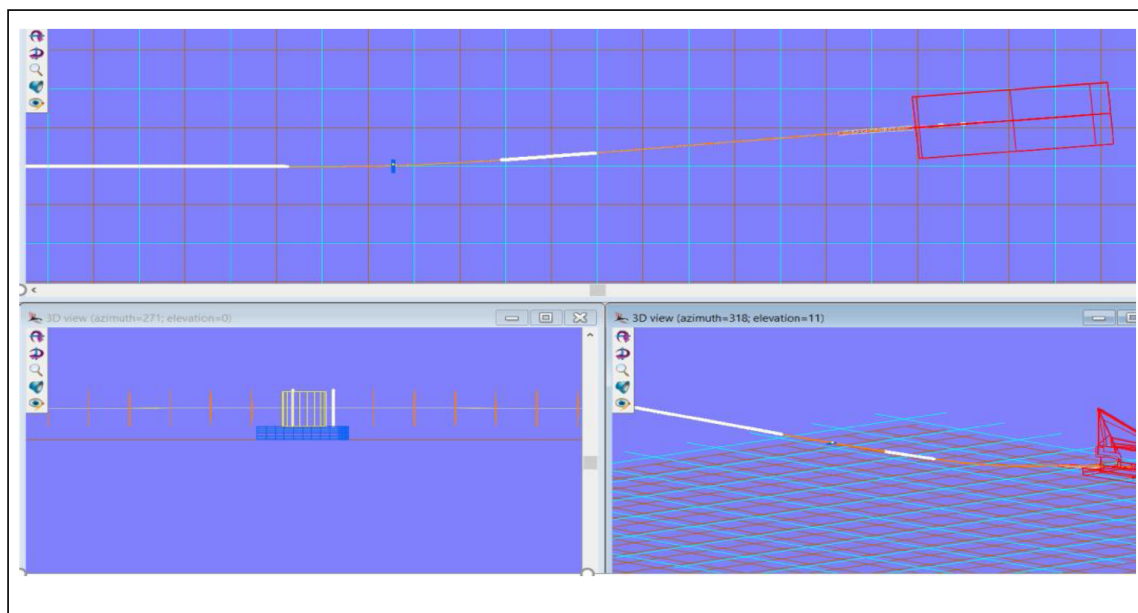


Figure 13—Typical Trunkline ZRB laying completion.

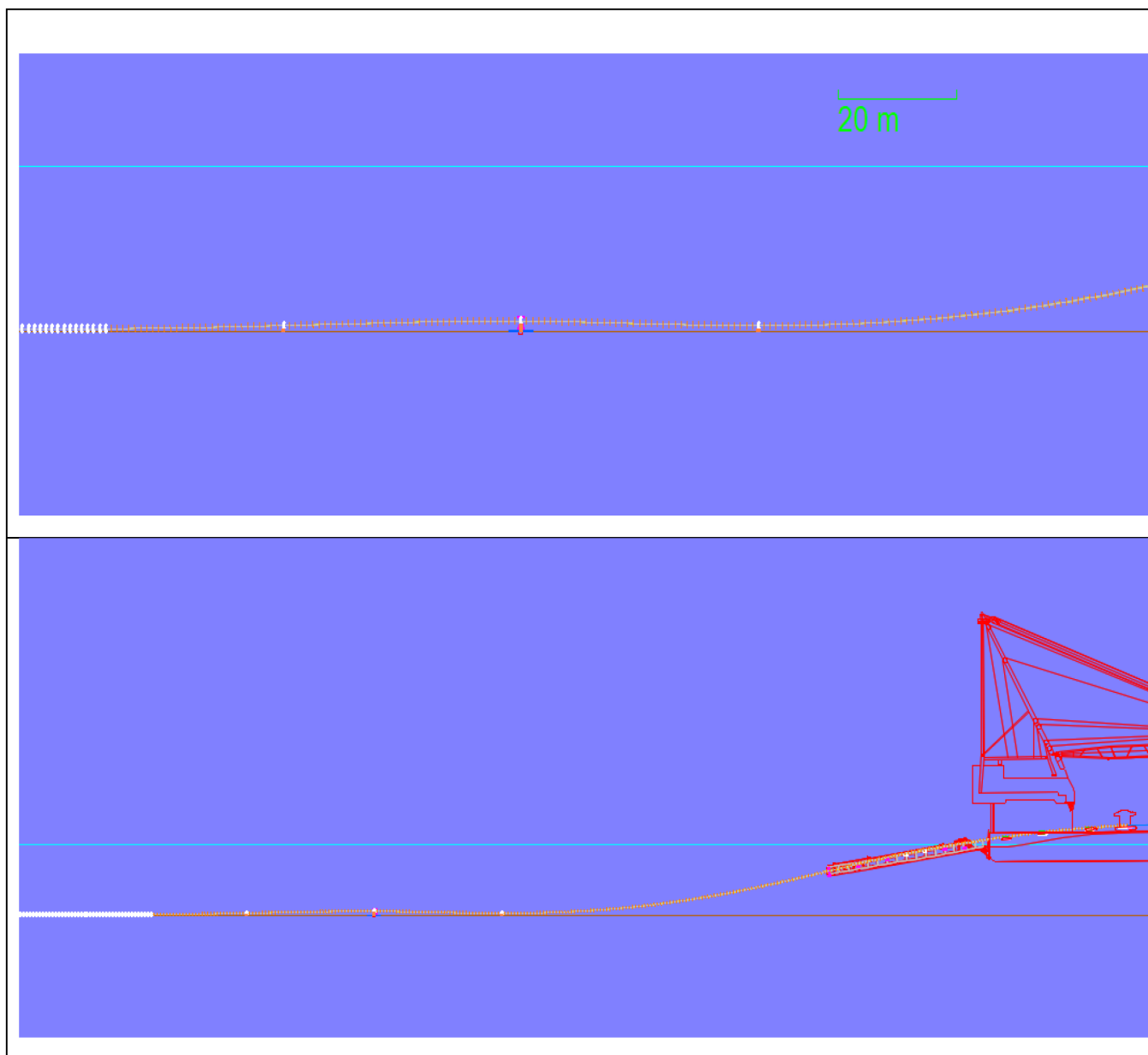


Figure 14—Typical Intra-field Pipeline TDP over 3 × ZRB Sleepers
(white dots represent contact between pipeline and seabed/sleeper)

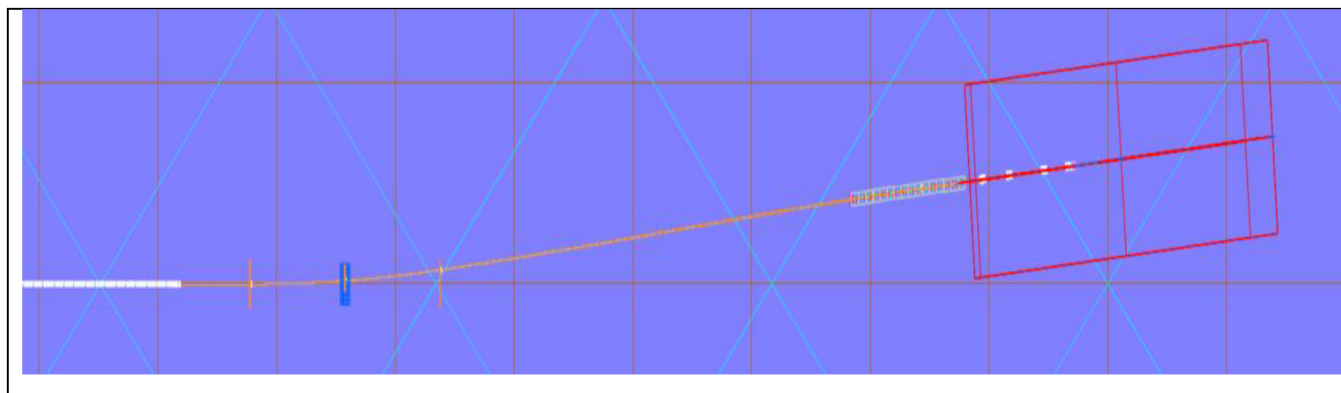


Figure 15—Typical Intra-field Pipeline plan view after change-of-heading over 3 × ZRB



Figure 16—Monitoring the Vessel Heading on the Ghost screen DB-30.



Figure 17—Monitoring the pipeline contact with the ZRB sleeper bollard with ROV

The summary results for cases (a), (b), (c), and (d), are presented in the tables below.

Table 5—Dynamic Installation Analysis Results for Rigid Stinger, 0% Structural Damping, and Nominal Change of Heading Angle of 5°

Pipeline [Empty Condition]	OD [mm]	WD [m]	Lay Tension [MT]	Change of Heading [°]	Stinger Type	Structural Damping [%]	Max. Sleeper Bollard Dyn. Load [MT]	Max. A&R Roller Dyn. Load [MT]
28"/25.4mm WT/40mmCWC	792.3	21	50	Nominal 5°	Rigid (Truss)	0%	14.8	11.4
		58	90				17.7	20.9
38"/26.97mmWT/40mmCWC	1046.3	18	35				18.7	23.9
		52	55				20.7	37.0

Table 6—Dynamic Installation Analysis Results for Rigid Stinger, 0% Structural Damping, and very Tight Curve Change of Heading Angle of 6°

Pipeline [Empty Condition]	OD [mm]	WD [m]	Lay Tension [MT]	Change of Heading [°]	Stinger Type	Structural Damping [%]	Max. Sleeper Bollard Dyn. Load [MT]	Max. A&R Roller Dyn. Load [mT]
28"/25.4mm WT/40mmCWC	792.3	21	50	Tight Curve 6°	Rigid (Truss)	0%	16.4	13.9
		58	90				19.5	18.4
38"/26.97mmWT/40mmCWC	1046.3	18	35				20.7	24.6
		52	55				21.6	36.5

Table 7—Dynamic Installation Analysis Results for Rigid Stinger, 5% Structural Damping, and Nominal Change of Heading Angle of 5°

Pipeline [Empty Condition]	OD [mm]	WD [m]	Lay Tension [MT]	Change of Heading [°]	Stinger Type	Structural Damping [%]	Max. Sleeper Bollard Dyn. Load [MT]	Max. A&R Roller Dyn. Load [MT]
28"/25.4mm WT/40mmCWC	792.3	21	50	Nominal 5°	Rigid (Truss)	5%	13.3	16.6
		58	90				15.3	13.6
38"/26.97mmWT/40mmCWC	1046.3	18	35				15.8	25.1
		52	55				16.1	27.0

Table 8—Dynamic Installation Analysis Results for Rigid Stinger, 5% Structural Damping, and very Tight Curve Change of Heading Angle of 6°

Pipeline [Empty Condition]	OD [mm]	WD [m]	Lay Tension [mT]	Change of Heading [°]	Stinger Type	Structural Damping [%]	Max. Sleeper Bollard Dyn. Load [mT]	Max. A&R Roller Dyn. Load [mT]
28"/25.4mm WT/40mmCWC	792.3	21	50	Tight Curve 6°	Rigid (Truss)	5%	14.8	15.9
		58	90				17.2	12.2
38"/26.97mmWT/40mmCWC	1046.3	18	35				17.1	28.3
		52	55				17.4	27.1

Table 09—Dynamic Installation Analysis Results Nominal Change of Heading Angle of 5°, Nominal Pipeline-Bollard Clearance of 0.5m, and Lateral Pipe-Seabed/Sleeper Friction Factor (LE / BE)

Pipeline in Empty Condition	OD [mm]	WD [m]	Lay Tension [mT]	Sleepers in the Lay direction			Sleeper Spacing Distance (m)	Best Est. Change of Heading [°]	Best Est. CL Lay Position ⁽¹⁾ [m]	Friction Sensitivity Cases	Seabed Soil Lateral Friction Factor (residual)	ZRB Sleeper Lateral Friction Factor	ZRB Bollard Sleeper			Max. Bollard Lateral Dynamic Reaction [mT]	Maximum Bollard Lateral Dynamic Reaction [mT]	Roller Dynamic Reaction Force (mT)	LCC			
[Friction Factor Sensitivities]				S2 Sleeper Height (m)	ZRB Sleeper Height (m)	S3 Sleeper Height (m)							RoC [m]	Projected Vertical RoC [m]	Projected Horizontal RoC [mT]				Barge [-]	Stinger [-]	Sag bend [-]	Bollard [-]
28"/25.4mm WT /40mmCWC	792.3	21	50	0.4	1	0.4	40	Nominal 5°	0.9	LE	0.75	0.3	445	649	612	20.1	10.5	7.9	0.28	0.28	0.35	0.38
										BE	0.97	0.65	455	681	612	20.1	10.2	8.0	0.28	0.28	0.35	0.38
										HE	1.35	0.7	456	683	612	20.1	10.2	8.0	0.28	0.28	0.35	0.38
		58	90							LE	0.62	0.3	416	578	600	20.0	13.3	12.1	0.67	0.75	0.40	0.46
										BE	0.86	0.65	424	598	600	20.0	13.3	12.2	0.66	0.76	0.40	0.47
										HE	1.12	0.7	425	602	600	20.0	13.4	12.1	0.67	0.74	0.40	0.47
38"/26.97mmWT /40mmCWC	1046.3	18	35	-	0.3	-	-		1	LE	0.75	0.3	844	960	1776	15.7	19.1	13.8	0.43	0.44	0.25	0.26
										BE	0.95	0.65	847	964	1776	15.7	19.8	14.1	0.43	0.43	0.25	0.26
										HE	1.25	0.7	839	952	1776	15.7	20.8	14.4	0.43	0.43	0.26	0.28
		52	55							LE	0.54	0.3	871	999	1779	15.9	16.9	36.1	0.71	0.62	0.41	0.43
										BE	0.9	0.65	854	973	1779	15.9	14.9	36.3	0.72	0.62	0.42	0.44
										HE	1.35	0.7	844	959	1779	15.9	15.0	36.3	0.70	0.61	0.43	0.44

Table 10—Dynamic Installation Analysis Results for Best Estimate Lateral Friction Factor, Nominal Pipeline-Bollard Clearance of 0.5m, and Sensitivity on the Pipeline Change-of-Heading Angle of 5° (-0.5°, +0.5°, +1.0°)

Pipeline in Empty Condition	OD [mm]	WD [m]	Lay Tension [mT]	Sleepers in the Lay direction			Sleeper Spacing Distance (m)	Sensitivity on Change of Heading	Change of Heading [°]	Best Est. CL Lay Position [m]	Best Est. Lateral Friction Factor (μ _{res})	Best Est. ZRB Sleeper Friction Factor (μ _{sleeper})	Minimum Radius of Curvature [m]	ZRB Bollard Sleeper				Max. Bollard Lateral Dynamic Reaction [mT]	Roller Dynamic Reaction Force (mT)	LCC			
[Change of Heading Angle Sensitivities]				S2 Sleeper Height (m)	ZRB Sleeper Height (m)	S3 Sleeper Height (m)								RoC [m]	Projected Vertical RoC [m]	Projected Horizontal RoC [mT]	Barge [-]			Stinger [-]	Sag bend [-]	Bollard [-]	
Intra-field Pipeline 28"/25.4mm WT/40mmCW C	792.3	21	50	0.4	1	0.4	45	Low	4.5	0.9	0.97	0.65	480	775	612	20.1	9.6	11.7	0.28	0.29	0.31	0.34	
								High	5.5				429	601	612	20.1	10.8	7.5	0.28	0.28	0.38	0.43	
								Exceed	6				405	539	612	20.1	11.6	6.3	0.28	0.28	0.43	0.48	
		58	90					Low	4.5				451	685	600	20.0	12.3	11.5	0.66	0.75	0.36	0.42	
								High	5.5				398	532	600	20.0	14.3	11.3	0.67	0.74	0.44	0.52	
								Exceed	6				374	478	600	20.0	15.4	11.2	0.65	0.73	0.48	0.57	
Trunklines 38"/26.97mm WT/40mmCW C	1046.3	18	35	-	0.3	-	-	Low	4.5	1	0.95		925	1083	1775	15.7	17.7	17.7	0.46	0.41	0.22	0.22	
								High	5.5				776	862	1777	15.7	20.6	19.1	0.47	0.44	0.29	0.31	
								Exceed	6				711	776	1777	15.7	20.5	15.7	0.44	0.42	0.32	0.34	
		52	55					Low	4.5				924	1081	1778	15.9	13.7	37.1	0.71	0.63	0.38	0.39	
								High	5.5				788	879	1780	15.9	16.1	35.3	0.74	0.62	0.46	0.48	
								Exceed	6				735	806	1782	15.9	17.2	35.2	0.74	0.61	0.50	0.53	

ZRB sleepers have advantages over the Vertical Upset Method

During the bid and FEED phases of the project, a decision was made to adopt the vertical upset method. However, after a thorough evaluation during the EPCI stage, it was concluded that ZRB sleepers were the more effective choice for buckle mitigation. Additionally, this option presents significant advantages in optimizing the offshore schedule and enhancing operational feasibility with DB-30.

The following table presents a thoughtful comparison of both methods, highlighting their key features and differences.

Table 11—Detailed comparison of the vertical offset vs ZRB

Mitigation Method	PROs	CONs
Vertical Upset	• Simplicity of pipeline routing • Straight Pipe-lay similar to crossings • Combination of vertical out-of-straightness and long raised section of pipe (from the seabed)	• Large Spans leading to snagging hazard and VIV susceptibility • Typically, lower reliability if not combined with other measures (CWC stripping, buoyancy modules); this is due to a large radius of curvature leading to a high critical buckling force at the trigger location. • Seabed features or unwanted HOOS locations compete with sleepers and increase the potential for uncontrolled and unforeseen buckling • High critical buckling force leads to the possibility of accumulated strain (and higher stress ranges) due to the formation of a single buckle rather than multiple buckles • Pipeline coating prone to damage, wear/tear due to cycles of shutdown and restart at the contact points with sleepers
ZRB	• High reliability of triggering a buckle at the intended location due to lower critical buckling force. • Reduces the chances of uncontrolled (natural) buckles forming away from the buckle trigger locations. • Allow closer buckle trigger spacing, which reduces the amount of thermal expansion/feed-in accommodated by each buckle and subsequently lowers maximum strain and stress ranges along the pipe on buckle section. • Reduced susceptibility to VIV or snagging hazard with a low-profile vertical trigger • Reduce overall installation risk compared to the ‘vertical sleeper’ method with fewer sleepers to install and lay pipe. • Reduced risk of coating damage to pipeline due to fewer sleepers used to achieve lateral buckle mitigation.	• Requirement for accuracy of installation, touchdown monitoring, etc. • Design of the ZRB structure to avoid pipeline sliding during pipe lay • Complicated routing to accommodate frequent changes of heading at the ZRB locations. • Additional length of pipeline will be required due to routing for pipeline heading changes at ZRB sites. However, this is generally not significant, being in the order of a few pipe joints for each line.

Operational Challenges

Despite the numerous advantages of ZRB sleepers, various operational challenges may arise during offshore pipeline installation. It is essential to ensure that the pipeline is installed correctly, and that proper contact is maintained between the pipeline and the ZRB bollard. Below is a list of some key challenges encountered during the execution stage.

- Large-diameter pipes with thick concrete weight coatings are at a high risk of cracking when they bend sharply at ZRB sleepers. Therefore, it is crucial to carefully plan the entire operation. This will help ensure smooth vessel operation and prevent the impact or shock loads that could damage the pipes.
- The derrick barge and its associated mooring must be meticulously designed and engineered to facilitate smooth vessel-heading changes. Specifically, these adjustments should allow the vessel to gradually transition its heading from 0 ° to 5 ° with precision.
- Continued monitoring of the touchdown points on the adjacent sleepers of the ZRB sleepers should be conducted to ensure that the pipeline laying is in the intended locations. This monitoring will be carried out by the Remotely Operated Vehicle (ROV)
- The lay tolerances for the pipeline were set at 0.5 m from the ZRB bollard. It is essential that the pipeline is laid on the ZRB within these tight tolerance limits. Achieving this precision involves significant operational challenges on-site, necessitating thorough engineering to ensure optimal performance.
- Constant tension must be maintained until the touchdown and heading changes are complete; otherwise, there is a high risk of the pipeline coming out of the bollard during the vessel's heading adjustments.

- The ZRB sleeper and its associated bollard must be engineered to effectively withstand side loads during changes in the vessel heading direction.
- The capacity of the vertical rollers on the barge stern is a crucial aspect that must be engineered with precision. As a vessel navigates, it encounters significant loads on the rollers, which must be designed to effectively endure these forces
- The entire operation should be performed in a controlled environment. Based on the pipelay and mooring engineering analyses, a suitable sea state was defined. In this case, the sea state was limited to a significant wave height (H_s) of 1.5 m.

Conclusion

In summary, the paper offers valuable insights into the use of ZRB sleepers for lateral buckling mitigation. The success demonstrated in this case study, drawn from a live EPCI project, highlights that the ZRB method effectively addresses lateral buckling issues for large-diameter pipes without compromising pipe integrity. However, it presents numerous operational challenges and must be meticulously planned and engineered to achieve optimal results. As operators increasingly pursue optimized designs and cost-effective solutions, the ZRB approach is likely to gain wider adoption in future subsea pipeline projects, as lateral buckling in the pipeline design can be effectively managed using this method.

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Abbreviations

DAF:	Dynamic Amplification Factor.
DB:	Derric Barge
DNV:	Det Norske Veritas
HP:	High Pressure
HT:	High Tension
H_s :	Significant Wave Height
RAOs:	Response Amplitude Operator
ROV:	Remotely Operated Vehicle
ZRB:	Zero Radius Bending.
EPCI:	Engineering Procurement, Construction, and Installation

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